

Actual BOM FINAL

Friday, January 30, 2026 3:51 PM

Block 1 — Input protection, rectification, and 5 V linear

Ref	Part Number	Description	Qty
J1	PJ-063AH	DC barrel jack, through-hole, center-positive	1
F1	OPTF0078P	Fuseholder	1
F2	0239003.MXP	3 A fuse, time-delay (use 0239003.HXP footprint)	1
D1	SBR3U60P1	3 A, 60 V SBR rectifier	1
C1	EDK476MD5DS9HAA	47 μ F electrolytic capacitor	1
C2	C1210C106K5PACTU	10 μ F, 1210, X5R ceramic capacitor	1
M1	LM340MPX-5.0/NOPB	5 V linear regulator (LM7805-type)	1
C3	C0603C104K5RACTU	0.1 μ F, 0603, X7R ceramic capacitor	1
C4	C0603C104K5RACTU	0.1 μ F, 0603, X7R ceramic capacitor	1
D2	TL431BQDBZR	Precision shunt regulator	1

Block 2 — VIN sense, comparator/EN/F2 logic network

Ref	Part Number	Description	Qty
C5	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor	1
R1	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (primary shunt sense)	1
R2	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor	1
R3	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor	1
R5	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor	1
R7	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor	1
R1_EN1	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (EN logic)	1
R1_EN2	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (EN logic)	1
R1_EN3	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (EN logic)	1
R1_F2	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (F2 logic)	1
R2_F2	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (F2 logic)	1
R4	CRCW060327K4FKEA	27.4 k Ω , 0603, 1% resistor	1
R6	CRCW060339K2FKEA	39.2 k Ω , 0603, 1% resistor	1

Block 3 — Logic switching, comparator, P-FET branch

Ref	Part Number	Description	Qty
C1_L1	C1210C226M8PACTU	22 μ F, 1210, X6S, 63 V ceramic capacitor	1

C2_L1	C1210C226M8PACTU	22 μ F, 1210, X6S, 63 V ceramic capacitor	1
L1	LM339BIRTER	Quad comparator, open-collector outputs	1
T1	FMB3904	NPN transistor, SOT-23	1
R7	CRCW0603100KFKEA	100 k Ω , 0603, 1% resistor (VIN_FUSED branch to P-FET source)	1
M2	SI2333CDS-T1-GE3	P-channel MOSFET, logic-level, SOT-23	1

Block 4 — Buck1 (TPS543321) + peripherals (updated R9)

Ref	Part Number	Description	Qty
C6	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (input bulk)	1
C7	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (input bypass)	1
U1	TPS543321RQFR	3 A synchronous buck regulator	1
L1	IHLP2525CZER2R2M01	2.2 μ H shielded inductor, 2525 footprint	1
R8	TNPV1206330KBYEA	330 k Ω , 1206, 0.1% resistor (MODE pin configuration)	1
Boot_C1	C0402C223K5RACTU	22 nF, 0402, X7R, 50 V ceramic capacitor (bootstrap)	1
R9	CRCW040256K2FKED	56.2 k Ω , 0402, 1% resistor (feedback top, raised to $\sim$ 5.3 V)	1
R10	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (feedback bottom)	1
C8	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (output bulk)	1
C9	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (output bypass)	1

Block 5 — Buck2 (TPS563300) + peripherals (updated R10)

Ref	Part Number	Description	Qty
C10	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (input bulk)	1
C11	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (input bypass)	1
U2	TPS563300DRLR	3 A synchronous buck regulator	1
L2	IHLP2525CZER2R2M01	2.2 μ H shielded inductor, 2525 footprint	1
Boot_C2	C0402C223K5RACTU	22 nF, 0402, X7R, 50 V ceramic capacitor (bootstrap)	1
R10	CRCW040256K2FKED	56.2 k Ω , 0402, 1% resistor (feedback top, raised to $\sim$ 5.3 V)	1
R11	CRCW060310K0FKEA	10 k Ω , 0603, 1% resistor (feedback bottom)	1
C12	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (output bulk)	1
C13	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (output bypass)	1

Block 6 — Buck3 (LT8609A) + peripherals (updated R12/R13)

Ref	Part Number	Description	Qty
C10	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (input bulk)	1

C11	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (input bypass)	1
U3	LT8609AEDDM#TRPBF	3 A, 2 MHz synchronous buck regulator	1
C12	C0805C105J5RACTU	1 μ F, 0805, C0G, 50 V ceramic capacitor (INTVcc decoupling)	1
R11	CRCW060318K2FKEA	18.2 k Ω , 0603, 1% resistor	1
Boot_C3	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (bootstrap)	1
L2	IHLP2525CZER2R2M01	2.2 μ H shielded inductor, 2525 footprint	1
R12	CRCW0603845KFKEA	845 k Ω , 0603, 1% resistor (feedback top, targets ~5.36 V)	1
R13	CRCW0603187KFKEA	187 k Ω , 0603, 1% resistor (feedback bottom)	1
C13	600S100JT250XT	10 pF, 250 V RF ceramic capacitor (compensation)	1
C14	C1210C226M5PACTU	22 μ F, 1210, X5R, 50 V ceramic capacitor (output bulk)	1
C15	C0603C104K5RACTU	0.1 μ F, 0603, X5R, 50 V ceramic capacitor (output bypass)	1

Block 7 — TPS_1 (Buck1 5V_OUT protection)

Ref	Part Number	Description	Qty
TPS_1	TPS1H200AQDGNRQ1	Smart high-side switch, 40 V, 200 m Ω	1
R_CL_1	CRCW06031K20FKEA	1.2 k Ω , 0603, 1% resistor (current-limit configuration)	1
C_DEL_1	VJ0805Y681KXACW1BC	680 pF, 0805, Y5V ceramic capacitor (delay/timing)	1
C16	VJ0603Y105KXACW1BC	1 μ F, 0603, Y5V ceramic capacitor (decoupling)	1
C17	VJ0603Y104KXXAC	0.1 μ F, 0603, X7R ceramic capacitor (bypass/filter)	1

Block 8 — TPS_2 (Buck2 5V_OUT protection)

Ref	Part Number	Description	Qty
TPS_2	TPS1H200AQDGNRQ1	Smart high-side switch, 40 V, 200 m Ω	1
R_CL_2	CRCW06031K20FKEA	1.2 k Ω , 0603, 1% resistor (current-limit configuration)	1
C_DEL_2	VJ0805Y681KXACW1BC	680 pF, 0805, Y5V ceramic capacitor (delay/timing)	1
C18	VJ0603Y105KXACW1BC	1 μ F, 0603, Y5V ceramic capacitor (decoupling)	1
C19	VJ0603Y104KXXAC	0.1 μ F, 0603, X7R ceramic capacitor (bypass/filter)	1

Block 9 — TPS_3 (Buck3 5V_OUT protection)

Ref	Part Number	Description	Qty
TPS_3	TPS1H200AQDGNRQ1	Smart high-side switch, 40 V, 200 m Ω	1
R_CL_3	CRCW06031K20FKEA	1.2 k Ω , 0603, 1% resistor (current-limit configuration)	1
C_DEL_3	VJ0805Y681KXACW1BC	680 pF, 0805, Y5V ceramic capacitor (delay/timing)	1
C20	VJ0603Y105KXACW1BC	1 μ F, 0603, Y5V ceramic capacitor (decoupling)	1

C21	VJ0603Y104KXXAC	0.1 μ F, 0603, X7R ceramic capacitor (bypass/filter)	1
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Block 10 — Shared 5V_OUT node

Ref	Part Number	Description	Qty
C_BULK_5V	T54EE476M050ESA100	47 μ F, 50 V polymer tantalum capacitor, 2917 (bulk output cap)	1
C_MF_5V	VJ0603Y105KXACW1BC	1.0 μ F, 16 V, X7R, 0603 ceramic capacitor (connector-side, mid-freq)	1
C_HF_5V	VJ0603Y104KXXAC	0.1 μ F, 25 V, X7R, 0603 ceramic capacitor (connector-side, high-freq)	1
D_TVS_5V	SMF5.0A	5 V TVS diode, SOD-123FL	1
J_USB_C	USB4125-GF-A	USB-C receptacle, mid-mount	1

Block 11 — External adapter plug + recommended 1 m cables

Ref	Part Number	Description	Qty
P_ADAPT	70235K607	Barrel plug for 12 V, 2 A adapter (mates with PJ-063AH)	1
C_USBCC_1M	CAB004BT1MWH	Belkin 1 m USB-C to USB-C cable, 60 W-class, 3 A-rated power conductors	1
C_USBA_C_1M	F2CU029BT1M-BLK	Belkin 1 m USB-A to USB-C cable, up to 3 A charging	1
C_USBA_L_1M	CAA001BT1MWH	Belkin 1 m USB-A to Lightning cable, MFi-class for iPhone/iPad	1